PTSD Symptom Connections after the Wenchuan Earthquake

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Abstract

This study examines relationships among 17 PTSD symptom ratings from a public survey of adults bereaved by the Wenchuan earthquake. The analysis uses 344 complete response profiles to compare rating distributions, estimate a regularized association network, and assess overall and cross-group connectivity. Avoidance of thoughts and activities forms the strongest edge. Hypervigilance has the highest overall strength, while a foreshortened future has the highest bridge strength. Resampling shows consistent selection of the strongest edges and recurring centrality patterns, with greater variation among weaker connections and closely ranked symptoms.

1 Introduction

PTSD questionnaires record trauma-related experiences such as disturbing memories, avoidance, sleep difficulties, and changes in emotion or alertness. Relationships among these ratings show which experiences tend to occur together and how symptoms connect across questionnaire domains. A network representation makes these patterns visible at the level of individual symptoms.

McNally et al. used reports from Wenchuan earthquake survivors to illustrate a network perspective on PTSD [1]. Building on this setting, the present analysis examines three questions: which symptom pairs have the strongest associations, which items have substantial connections across symptom groups, and how consistently these patterns appear when the sample and estimation settings change.

The analysis combines rating distributions, pairwise correlations, and regularized partial associations. Bootstrap and sample-removal comparisons assess the stability of the estimated structure, while sensitivity analyses examine the effects of alternative estimation settings.

2 Data Summary

The public matrix contains 362 Chinese adults who survived the 2008 Wenchuan earthquake and lost at least one child. The `bgms` documentation identifies the instrument as the civilian PTSD Checklist for DSM-IV and links to the original CSV [2]. Its 17 items use ratings from 1 to 5, describing how much each symptom bothered the respondent during the preceding month. The questionnaire's theoretical total-score range is 17–85 [3].

Table 1 summarizes the data checks. The analysis excludes 18 profiles with missing ratings and retains 344 complete profiles without imputation. The retained data include two repeated complete response profiles. The file contains symptom ratings without participant identifiers, demographic variables, or repeated assessments.

Table 1: Source-data checks and the analysis sample.

Measure	Count
Source participants	362
Symptom items	17
Missing ratings	22
Incomplete profiles	18
Complete profiles retained	344
Invalid observed ratings	0
Repeated complete profiles after the first	2

3 Methods

3.1 Ratings and symptom groups

The descriptive analysis summarizes item ratings and total scores. Spearman correlations measure pairwise monotonic associations. Table 2 defines the figure labels and the three groups used to calculate bridge strength. These groups follow the DSM-IV questionnaire ordering [3].

Table 2: Item codes, abbreviated symptom descriptions, and predefined groups.

Code	Symptom	Group
`intrusion`	Intrusive memories	Re-experiencing
`dreams`	Distressing dreams	Re-experiencing
`flash`	Flashbacks	Re-experiencing
`upset`	Distress at reminders	Re-experiencing
`physior`	Physical reactions	Re-experiencing
`avoidth`	Avoidance of thoughts	Avoidance/numbing
`avoidact`	Avoidance of activities	Avoidance/numbing
`amnesia`	Trauma-related amnesia	Avoidance/numbing
`lossint`	Loss of interest	Avoidance/numbing
`distant`	Feeling distant	Avoidance/numbing
`numb`	Emotional numbness	Avoidance/numbing
`future`	Foreshortened future	Avoidance/numbing
`sleep`	Sleep difficulties	Hyperarousal
`anger`	Irritability/anger	Hyperarousal
`concen`	Concentration difficulties	Hyperarousal
`hyper`	Hypervigilance	Hyperarousal
`startle`	Exaggerated startle	Hyperarousal

3.2 Regularized association network

The network uses a continuous-score approximation to the ordinal ratings. For participant i and item j, the transformation assigns the normal score $z_{ij} = \Phi^{-1}((r_{ij} - 0.5)/n)$, where r_{ij} is the average rank among tied observations, $n = 344$, and Φ is the standard normal distribution function. Each transformed column then receives mean zero and variance one. This transformation preserves rating order and ties.

The graphical lasso estimates a positive-definite precision matrix Θ from the transformed-score covariance S by minimizing

$$\operatorname{tr}(S\Theta) - \log\det(\Theta) + \lambda \sum_{j \neq k} |\Theta_{jk}|. \tag{1}$$

The analysis compares 28 logarithmically spaced penalties from $\lambda_{\max}$ to $0.01\lambda_{\max}$, where $\lambda_{\max}$ is the largest absolute off-diagonal covariance. The selection criterion follows the EBIC form [4]:

$$\mathrm{EBIC} = n\{\operatorname{tr}(S\Theta) - \log\det(\Theta)\} + m \log n + 4\gamma m \log p, \tag{2}$$

where m counts nonzero undirected edges, $p = 17$, and the main setting uses $\gamma = 0.5$. The criterion is evaluated at the penalized precision-matrix estimates. The edge weight is $w_{jk} = -\Theta_{jk}/\sqrt{\Theta_{jj}\Theta_{kk}}$. These weights describe partial associations among transformed ratings after accounting for the remaining items.

3.3 Connectivity and stability

Overall strength is $s_j = \sum_{k \neq j} |w_{jk}|$. Bridge strength restricts this sum to items in other predefined groups [5]. The analysis re-estimates the transformation and penalty in 300 participant bootstrap samples and in 100 subsamples at each of four removal levels. These comparisons assess edge variation, top-three centrality membership, and correlations with full-sample centrality [6]. Bootstrap percentile ranges and selection frequencies summarize variation across the re-estimated networks. Additional models vary γ and use standardized raw ratings.

4 Findings

4.1 Symptom ratings and pairwise associations

The complete-profile total rating has mean 45.45 and standard deviation 14.58. Its median is 43, its interquartile range is 33–57, and observed totals range from 18 to 85.

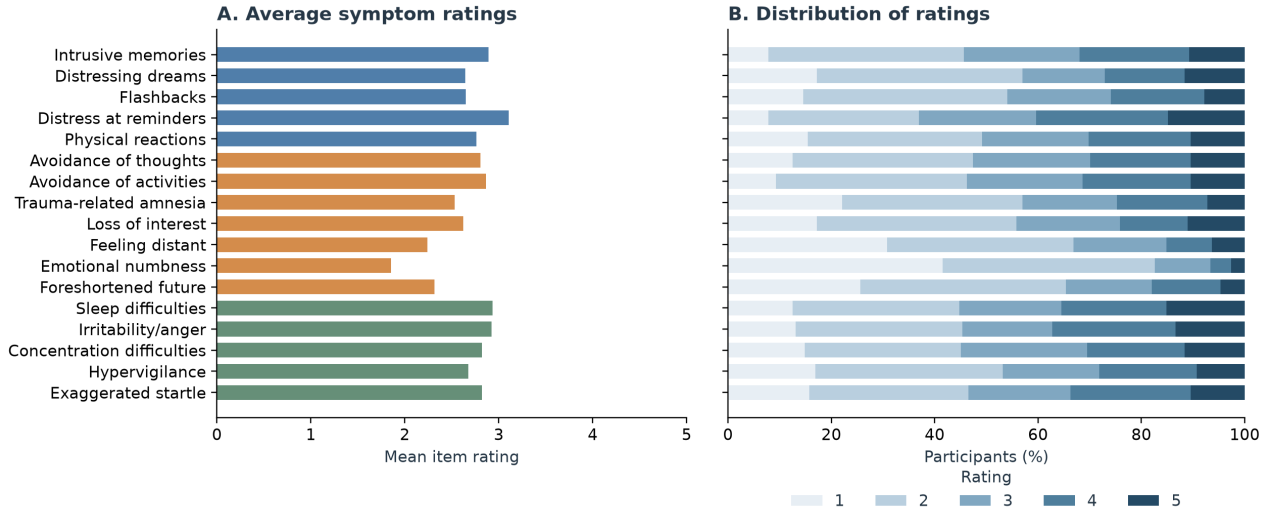


Figure 1: Mean symptom ratings and the distribution of all five response categories among 344 participants. Bar colors in panel A identify the three predefined symptom groups.

Figure 1 shows that distress at reminders has the highest mean rating (3.10), followed by sleep difficulties (2.93) and irritability/anger (2.92). Emotional numbness has the lowest mean (1.85). Ratings of at least 3 occur in 63.1% of responses for distress at reminders and 17.4% for emotional numbness.

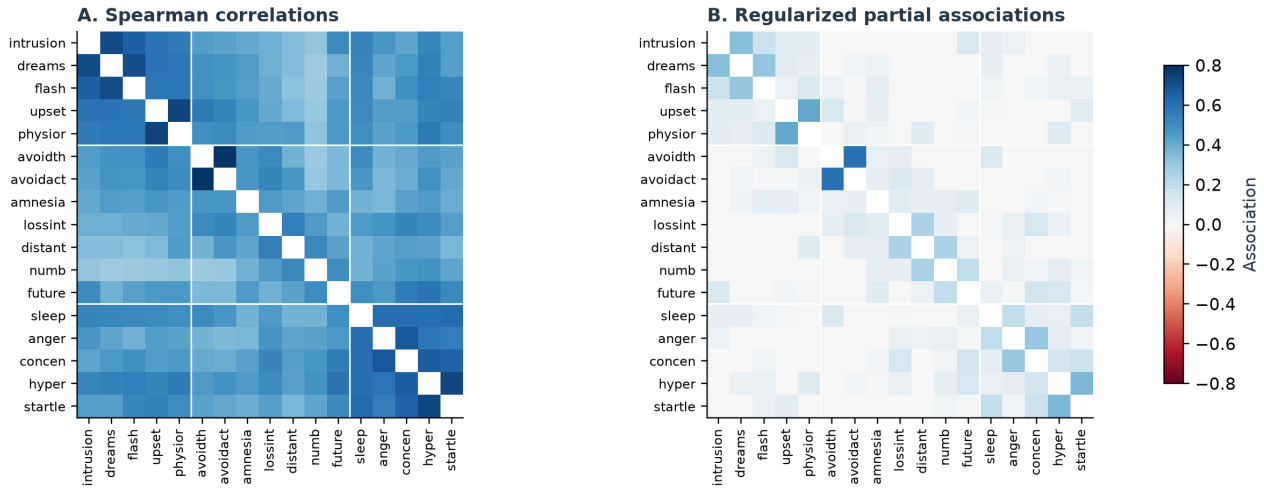


Figure 2: Pairwise Spearman correlations and regularized partial associations. Both panels use the same color scale; white separators mark the predefined groups. The diagonal is omitted.

All 136 pairwise Spearman correlations are positive, ranging from 0.290 to 0.787 with median 0.461. Figure 2 shows broad co-variation across the questionnaire. The partial-association panel contains smaller values and many zero weights: accounting for the other ratings and applying regularization yields a more selective set of connections.

4.2 The symptom network

The main model selects $\lambda = 0.043795$ and retains 82 of 136 possible edges, all positive. Forty-seven edges cross the predefined groups. The selected penalty lies inside the candidate range, and all 28 candidate fits converge.

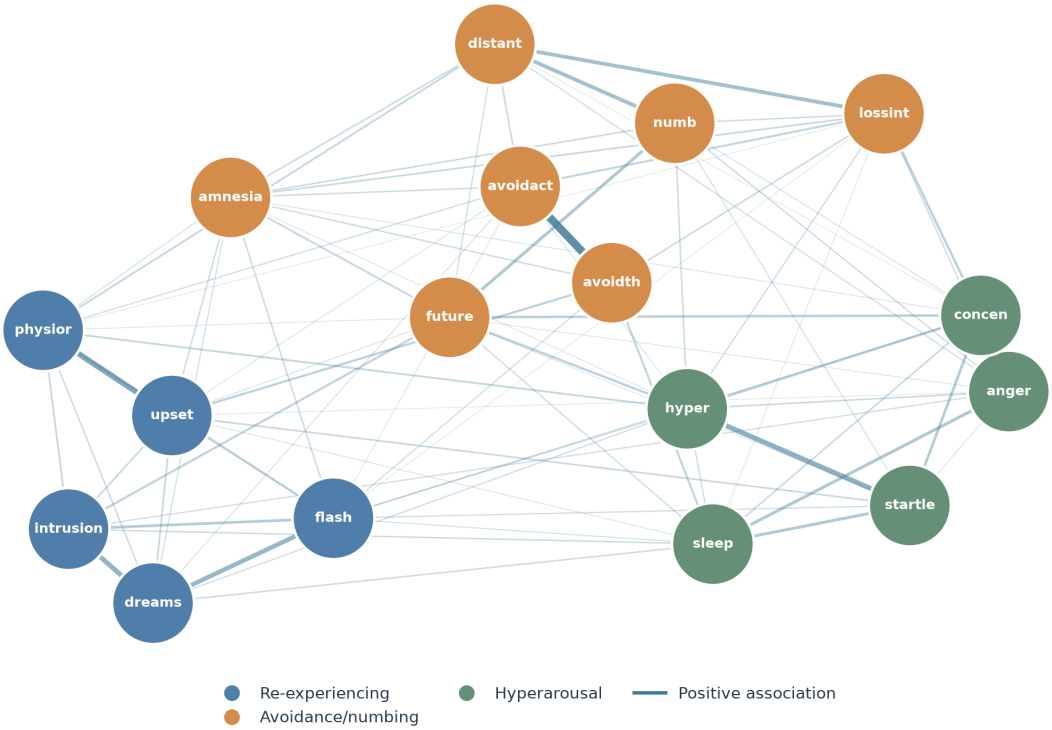


Figure 3: The regularized symptom network. Node colors indicate predefined groups, and darker, thicker lines indicate larger positive partial associations.

Table 3: The eight largest absolute edge weights. All eight connect symptoms within the same predefined group. Ranges contain the central 95% of bootstrap estimates.

Symptom pair	Weight	Bootstrap range
Avoidance of thoughts / activities	0.598	[0.497, 0.672]
Distress at reminders / physical reactions	0.412	[0.293, 0.497]
Hypervigilance / exaggerated startle	0.359	[0.237, 0.464]
Intrusive memories / distressing dreams	0.341	[0.211, 0.459]
Distressing dreams / flashbacks	0.311	[0.212, 0.407]
Irritability/anger / concentration difficulties	0.306	[0.186, 0.401]
Feeling distant / emotional numbness	0.254	[0.162, 0.354]
Loss of interest / feeling distant	0.253	[0.145, 0.337]

Figure 3 and Table 3 identify particularly strong connections within avoidance, reminder-related responses, and alertness-related symptoms. The largest cross-group edge connects a foreshortened future with concentration difficulties (0.148). The many smaller cross-group edges link symptoms across the predefined domains, complementing the stronger within-group associations.

4.3 Overall connectivity and bridges

Figure 4 compares two aspects of connectivity. Hypervigilance has the highest overall strength (1.149), followed by concentration difficulties (1.079) and avoidance of thoughts (1.026). A foreshortened future has the highest bridge strength (0.525), followed by hypervigilance (0.507) and concentration difficulties (0.375).

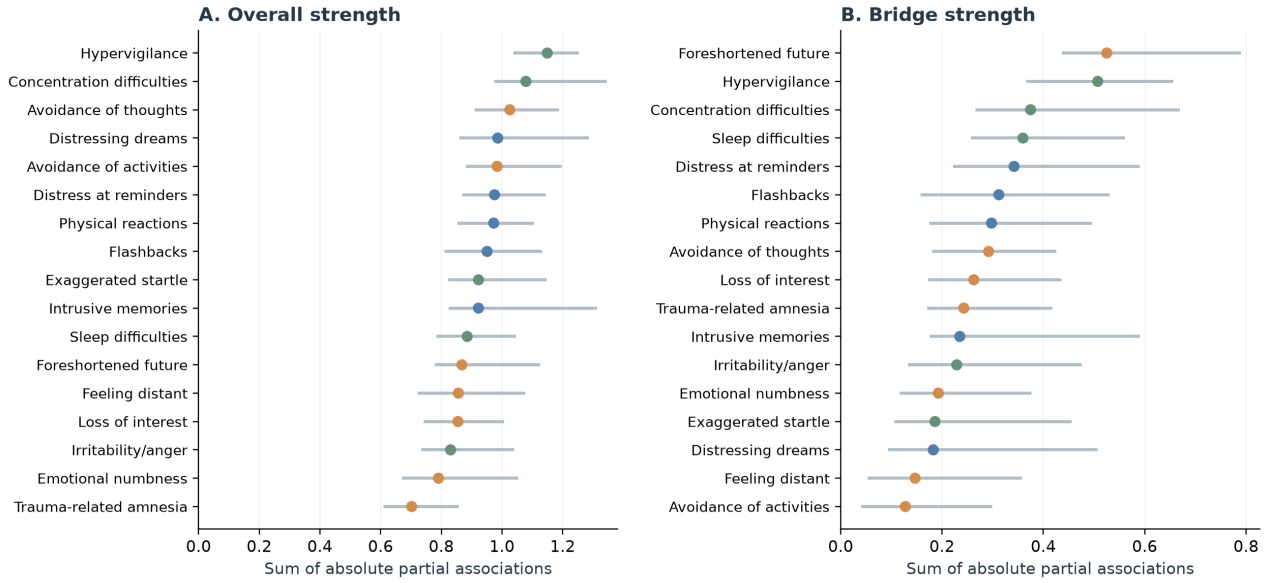


Figure 4: Overall strength and bridge strength for all 17 items. Points show full-sample estimates; horizontal lines contain the central 95% of bootstrap estimates. Each panel orders items by its own point estimates.

Table 4: The six items with the highest full-sample bridge strength. The final columns give bootstrap top-three membership for each centrality measure.

			Top-three membership (%)	
Symptom	Strength	Bridge	Strength	Bridge
Foreshortened future	0.868	0.525	1.3	95.3
Hypervigilance	1.149	0.507	84.0	80.3
Concentration difficulties	1.079	0.375	77.3	48.0
Sleep difficulties	0.885	0.360	1.3	24.3
Distress at reminders	0.975	0.342	12.3	19.7
Flashbacks	0.951	0.312	10.3	8.7

The two measures highlight different structures. Avoidance of thoughts gains substantial overall strength from its strong connection with avoidance of activities. A foreshortened future has lower overall strength but distributes a larger share of its connections across group boundaries. Its strongest cross-group connections include concentration difficulties (0.148), hypervigilance (0.136), and intrusive memories (0.120).

Table 4 shows how often these items appear among the three most connected symptoms across bootstrap samples. A foreshortened future belongs to the bridge-strength top three in 95.3% of samples, and hypervigilance does so in 80.3%. Concentration difficulties appears in the bridge-strength top three in 48.0% of samples. Hypervigilance and concentration difficulties also frequently occupy the overall-strength top three.

The bootstrap results show recurring patterns of overall and cross-group connectivity. Overlapping bootstrap ranges in Figure 4 indicate uncertainty in the exact ordering of items with similar centrality estimates.

4.4 Resampling uncertainty

All 300 bootstrap samples produce successful fits. Figure 5 shows the ten largest full-sample edges. Every one of these edges appears with the same positive sign in all 300 samples. The avoidance-of-thoughts/activities edge has a bootstrap range of 0.497–0.672, while the distress-at-reminders/physical-reactions edge has a range of 0.293–0.497.

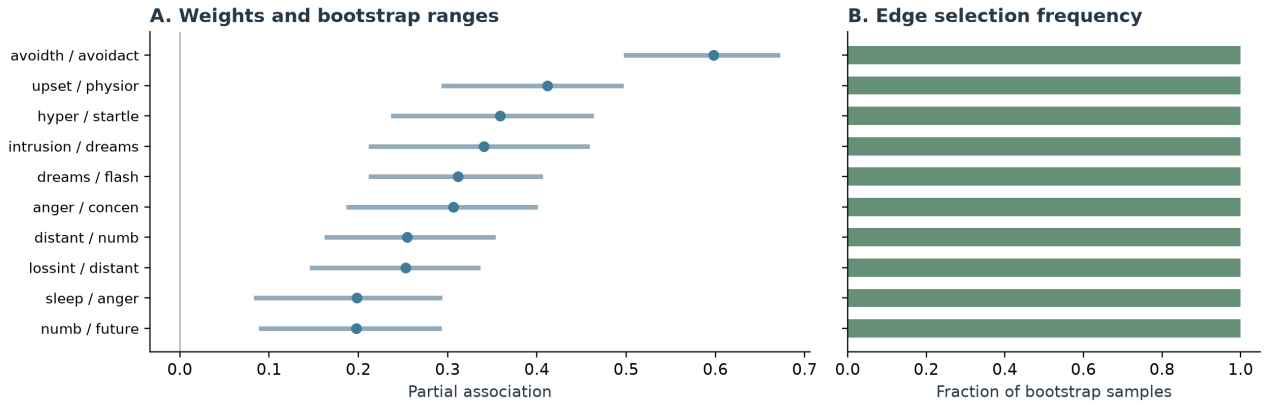


Figure 5: Bootstrap variation for the ten largest full-sample edges. Points show original weights, lines show the central 95% of bootstrap estimates, and bars show nonzero-edge selection frequencies.

Weaker edges have more variable selection frequencies. Of the 82 edges selected in the main model, 57 appear in fewer than 95% of bootstrap samples and five appear in fewer than half. The strongest associations therefore recur more consistently across samples.

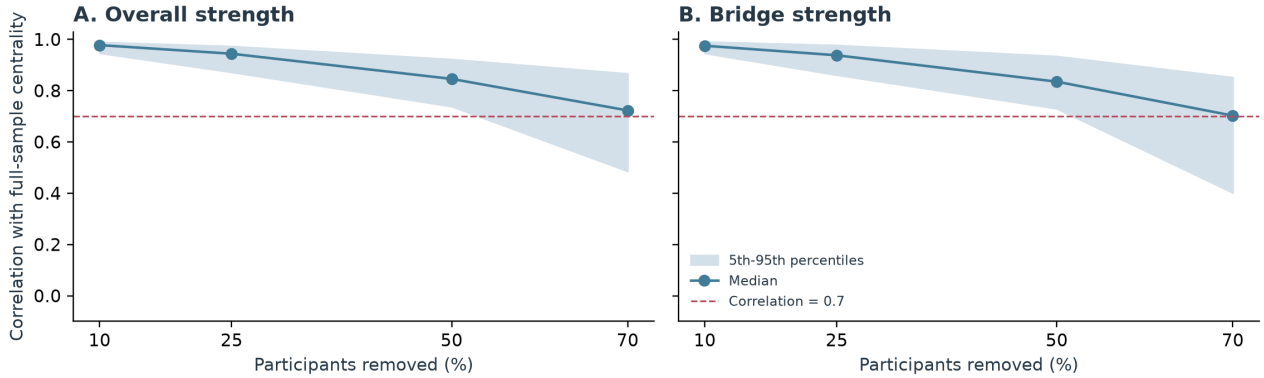


Figure 6: Centrality stability when participants are removed without replacement. Each level uses 100 subsamples. Lines show median correlations with full-sample centrality; shaded bands show the 5th–95th percentiles.

Figure 6 shows a gradual loss of stability as the sample shrinks. Removing 50% of participants retains median correlations of 0.846 for overall strength and 0.835 for bridge strength. Correlations reach at least 0.7 in 99% and 97% of these subsamples, respectively. At 70% removal, these proportions fall to 64% and 51%.

Both centrality vectors achieve a stability bound of 0.50 on the tested grid: they maintain correlation of at least 0.7 in at least 95% of subsamples at that level and every smaller tested level. All 400 subsample fits succeed, and no candidate penalties are excluded for nonconvergence in either resampling experiment.

4.5 Sensitivity to estimation settings

Figure 7 and Table 5 show that $\gamma \in \{0, 0.25, 0.50, 0.75\}$ selects the same rank-based model, with identical penalties, 82 edges, and unchanged centrality rankings.

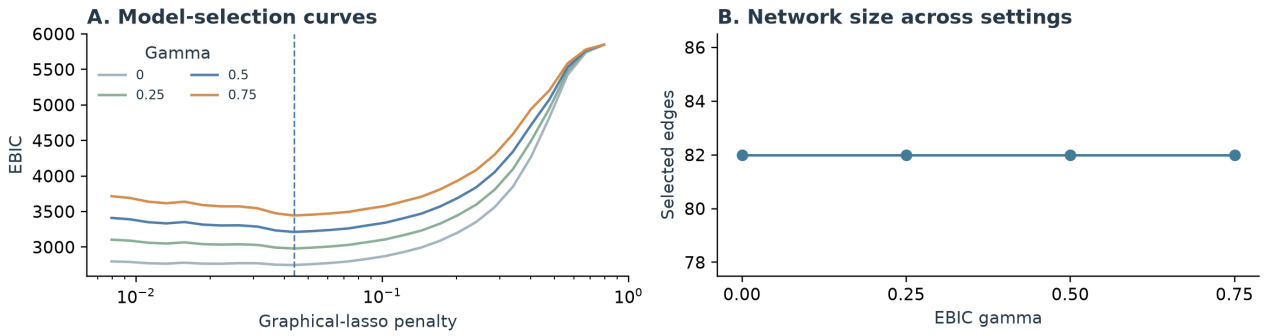


Figure 7: Model-selection curves and selected network sizes across EBIC settings. The vertical line marks the main selected penalty. All four settings select the same rank-based network.

Table 5: Sensitivity comparisons with the main rank-based model. Jaccard similarity compares selected edge sets; the final columns give Spearman correlations between centrality rankings.

Scores	γ	Edges	Jaccard	Strength ranks	Bridge ranks
Normal ranks	0.00	82	1.000	1.000	1.000
Normal ranks	0.25	82	1.000	1.000	1.000
Normal ranks	0.50	82	1.000	1.000	1.000
Normal ranks	0.75	82	1.000	1.000	1.000
Raw ratings	0.50	84	0.953	0.983	0.985

Standardized raw ratings yield 84 edges, including two negative edges, with edge-set similarity of 0.953 and edge-weight correlation of 0.995. Strength and bridge-strength rank correlations are 0.983 and 0.985. Hypervigilance and a foreshortened future retain the highest overall and bridge strength, respectively, in every tested setting.

5 Conclusion

The symptom network combines strong within-group associations with smaller connections across groups. Avoidance of thoughts and activities forms the strongest pair, hypervigilance has the highest overall strength, and a foreshortened future has the highest bridge strength. The ten strongest edges recur in all bootstrap samples, and the leading overall- and bridge-strength items remain unchanged across the tested estimation settings. These findings distinguish strong within-group connections from connectivity distributed across symptom groups, adding structural information to the symptom-rating summaries.

References

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